

Information Computer Systems and Python Programming

Midterm

Part I. Excel

Instructions:

You should find the data files for the exam in the D://Exam_Data folder on your computers, you should create a copy of the file and from then on work in this copy. Rename the copy according to a template

<Name>_<Surname>.xlsx. Each task (including the bonus ones) must be completed on a separate list in the file named <Task>_<number of the task>. The excel file with your solutions should be uploaded to the folder D://Exam_solution on your computers.

During the exam you will have an opportunity to earn bonus points via solving the bonus problems, note that **your total score for the exam won't exceed 100 points even if you score more than that through solving bonus problems.** The excel part overall can give you 40 points and additional 5 if you solve the bonus tasks.

Dataset card

Index – index of the stock on the market

Date – date of recorded stock prices

Open – opening price of the stock

High – highest stock price for the trading date

Low – lowest stock price of the trading date

Close – closing price of the stock

Adj close – closing price adjusted to corporate actions

Volume – trading volume of the stock

Task 1 (8 pts)

You want to take a loan of 1 000 000 rubles. The compounds every month. The bank offers you an interest rate of 12% on your loan if your monthly payment is 50 000 or a special rate to close your loan in 2 years if your monthly payment is 70 000. You are interested in paying off the loan as soon as possible.

In cells A1:C6 create the table, where in the first column you put the names of the characteristics of the offer and in the second and third columns you put the values for the characteristics according to each offer, name the second and third columns **Offer 1** and **Offer 2**. Calculate the missing characteristics via excel' financial functions. In cell E2 put the monthly payment of the most optimal offer for you.

Task 2 (8 pts)

In the A column of the list create the column that represents the absolute difference between the opening and the closing price of the stock. Then find the scalar **x**, by which the sum of the values in this new column should be scaled so that the absolute difference between this scaled sum and the function below is zero. The domain of **x** is (0, 10).

$$f(x) = \frac{\sqrt{\exp(\beta x)}}{\gamma}$$

Assume $\beta=15$ and $\gamma=100$

Put the resulting value of **x** in the cell C3. After that in the cells D5:H9 create a two-way table, where there'll be shown possible values of the **f(x)** defined above for $\beta= 0,5; 1; 5; 10$ and $\gamma=90; 60; 50; 30$.

The rest of the file you can use for the intermediate computations.

Task 3 (8 pts)

You have been told that the entries, which have the zero difference between the opening and closing price of the stock should be classified as EMPTY, while other entries should be classified as FULL. Also the entries where the closing price is over 2 percent lower than the opening price should be classified as VOLATILE, while the entries where the closing price is over 2 percent higher should be classified as PROMISING, the rest should be classified as CALM.

In the columns A through H copy the initial database, then in the I and J columns put the EMPTY/ FULL classes and the VOLATILE/ PROMISING/ CALM classes respectively. Call them class 1 and 2 respectively.

In the cells D1:E9 create a frequency table for the opening stock price, which would have the following values for the upper boundaries: 1000, 1500, 2000, 3000, 4000, 5000, 10000, 20000

Task 4 (8 pts)

Give the descriptive statistic (minimum, maximum, mean) for the highest stock prices per trading day given the following criteria:

1. The stock prices considered are larger than 4000
2. Their volume is not zero

Use the Database functions for this task. Put the values in cells B1:B3 while defining their nature (minimum/ maximum or mean) in the cells A1:A3.

Task 5 (8 pts)

Create a pivot table with the data from the 3rd task. Have class 1 values as columns, class 2 values as rows, while having Index and Date values as filters. Find the total volume of index GDAXI for the dates when it was a PROMISING stock. Put the result in cell D1.

Bonus task 1 (2 pts)

Do the classification that's required in task 3 without recursively calling any function. Put the resulting classes in columns A and B respectively.

Bonus task 2 (3 pts)

Build a histogram of the frequency table from task 3 (you can reconstruct it here as well)